

NWise Student-Led Cybersecurity Internship

WEEK 6 - DIGITAL FORENSICS

Professional Digital Forensic Investigation

Report

Report Field	Details
Case Reference	NWise_DF001
Investigator	Sukhmandeep Kaur
Organisation	NWise Cybersecurity
Department	Digital Forensics
Evidence	Digital Corpora nps-2009-ntfs1 / ntfs1-gen2.E01
Forensic Tool	Autopsy 4.23.1
Report Version	1.1 - Integrity Verified

This report documents the controlled Week 6 forensic investigation and records only findings supported by the examined evidence.

1. Investigator Details

Field	Details
Investigator Name	Sukhmandeep Kaur
Investigator ID	Not provided
Organisation	N Wise Cybersecurity
Department	Digital Forensics
Date of Investigation	11 August 2026
Report Version	1.1

2. Case Information

Field	Details
Case Reference	N Wise_DF001
Case Title	Week 6 - Digital Forensic Examination of nps-2009-ntfs1
Incident Type	Educational forensic investigation following reports of suspicious activity
Evidence Reference	Digital Corpora nps-2009-ntfs1; examined image: ntfs1-gen2.E01
Priority	Training / educational investigation
Case Status	Completed for the activities specified in the supplied lab documents

3. Investigation Objective

The objective was to examine a forensic copy of storage evidence supplied through Digital Corpora while preserving the original evidence. The investigation verified evidence integrity, imported the forensic image into Autopsy, examined the available NTFS structures, checked for deleted content, performed the required keyword searches, reviewed available activity artefacts, and exported evidence of interest.

4. Evidence Summary

Evidence ID	Description	Source	Acquisition Method
DF001-E01	ntfs1-gen2.E01 forensic disk image	Digital Corpora - nps-2009-ntfs1 dataset	Downloaded and preserved in 01_Original_Evidence; copied to 02_Working_Copy for analysis
DF001-E02	EFS-key-info.txt	ntfs1-gen2.E01 via Autopsy	Identified by keyword search and extracted to 04_Exports

5. Evidence Integrity Verification

Field	Value
Evidence File	ntfs1-gen2.E01
Hash Algorithm	SHA-256
Original Hash Value	2BADEAD91BEF56C80155D7731671AD1D93C08F3 2CD4CE17566FDF02D5769FEEA
Verified / Working Copy Hash Value	2BADEAD91BEF56C80155D7731671AD1D93C08F3 2CD4CE17566FDF02D5769FEEA
Integrity Verified	YES - hashes match exactly

PowerShell Get-FileHash was used with SHA-256 on both the preserved original and the working copy. The values matched exactly, demonstrating that the working copy used for analysis was identical to the preserved original at the time of verification.

6. Investigation Activity Log

Time	Investigation Activity	Result
Not recorded	Prepared NWise_DF001 workspace	Separate folders established for original evidence, working copy, Autopsy case, exports, notes and report.
Not recorded	Acquired nps-2009-ntfs1 dataset	Evidence downloaded from Digital Corpora.
Not recorded	Created working copy	ntfs1-gen2.E01 copied to 02_Working_Copy.
Not recorded	Verified SHA-256 integrity	Original and working-copy hashes matched exactly.
Not recorded	Created Autopsy case	NWise_DF001 created; ntfs1-gen2.E01 added as Disk Image or VM File.
Not recorded	Examined file-system structure	Observed \$Extend, \$Unalloc, Compressed, Encrypted, RAW and System Volume Information.
Not recorded	Reviewed Deleted Files	No recoverable deleted files identified.
Not recorded	Performed keyword searches	password=6; confidential=4; finance=2; invoice=0.
Not recorded	Reviewed activity/artifact results	No dedicated Recent Activity category generated.
Not recorded	Exported relevant evidence	EFS-key-info.txt exported to 04_Exports.

7. Digital Evidence Identified

Evidence	Location	Reason for Interest
EFS-key-info.txt	Autopsy keyword results; exported to 04_Exports	Contained text stating that the password for the EFS key is "password"; relevant to encrypted-file access.
EFS-key-password-strong-protection.pfx	Password keyword results	EFS-related certificate/key artifact associated with password hits.
report02-3.pdf	Compressed and RAW areas	Produced hits for password, confidential and finance.
NISTSP800-88_rev1.pdf	Compressed and RAW areas	Produced confidential keyword hits.

8. Deleted Files Recovered

File Name	Original Location	Recovery Status
None identified	N/A	Deleted Files category showed no recoverable deleted files for the examined image.

9. Keyword Search Results

Keyword	Matches	Notes
password	6	Notable hit: EFS-key-info.txt stated that the EFS key password was "password". Other hits included report02-3.pdf, \$MFT and EFS-related PFX artifacts.
confidential	4	Hits associated with NISTSP800-88_rev1.pdf and report02-3.pdf in Compressed and RAW locations.
finance	2	Both hits associated with report02-3.pdf; preview referenced the 1997 Economic Census, Finance and Insurance.
invoice	0	No matching evidence identified.

10. Timeline Observations

A dedicated Recent Activity category was not generated from ntfs1-gen2.E01 in the observed Autopsy results. Therefore, this examination did not establish a reliable timeline of recently opened files, browser history, downloads, installed applications, user logins, or connected USB devices. Other indexed artefacts

visible in Autopsy included keyword hits, email addresses, IP addresses, phone numbers, URLs, metadata, and communication-account information.

11. Findings

Autopsy successfully processed ntfs1-gen2.E01. The available structure included \$Extend, \$Unalloc, Compressed, Encrypted, RAW and System Volume Information. The standard Windows folders referenced by the lab, including Users, Windows, Program Files and ProgramData, were not present in the examined image, which limited user-profile analysis.

No recoverable deleted files were identified. The most significant keyword finding was EFS-key-info.txt, which contained a plaintext statement that the EFS key password was "password". The file was exported to 04_Exports. Additional keyword hits were documented in report02-3.pdf, NISTSP800-88_rev1.pdf, \$MFT and EFS-related PFX material. The finance and confidential matches were recorded without treating the keyword matches alone as evidence of malicious activity.

The original and working-copy SHA-256 hashes matched exactly, confirming the integrity of the working forensic image used for analysis.

12. Conclusion

The investigation objective was achieved for the evidence available in ntfs1-gen2.E01. The original evidence was preserved, a working copy was created, and SHA-256 verification confirmed that both copies matched. The image was imported into Autopsy and examined using a structured forensic workflow. Keyword searches identified EFS-related evidence, including a plaintext reference to an EFS key password, and EFS-key-info.txt was exported for preservation. No deleted files were recovered and no dedicated Recent Activity results were available from this image.

13. Recommendations

- Preserve the original forensic image and exported evidence without modification.
- Retain both SHA-256 outputs with the case notes as integrity-verification evidence.
- Perform additional examination of EFS-related certificate/key artifacts if the investigation scope is extended.
- Review indexed URLs, email addresses, metadata and other artefacts only where relevant to a defined investigative question.
- Retain screenshots, hash outputs, exported evidence and investigation notes with the final report to support repeatability.

14. Investigator Declaration

I certify that this investigation was conducted using accepted digital forensic principles. To the best of my knowledge, the evidence was handled in a manner that preserved its integrity and the findings presented in this report accurately reflect the results of my examination.

Declaration Field	Entry
Investigator Signature	

Date	_____

Final Review Checklist

Checklist Item	Status
Evidence integrity verified using SHA-256	YES
Investigation activity log completed	YES
Key evidence documented	YES
Relevant artefacts exported	YES - EFS-key-info.txt
Findings and conclusion completed	YES
Report reviewed before submission	YES - content prepared; investigator should perform final personal review/signature